

Companion XIX: Non-Linear Structure Formation in the Relaxation-Driven Cyclic Cosmology Halo Collapse, Cosmic Web Morphology, and Relaxation-Modified Dynamics

Michael Lehmann¹

¹Independent Researcher, Relaxation-Driven Cyclic Cosmology (RDCC) Framework,
mi.lehmann@gmx.de

May 15, 2026

Abstract

Non-linear structure formation provides a powerful probe of cosmological physics beyond the linear regime. Halo collapse, the cosmic web, and the distribution of non-linear density fluctuations encode information about the underlying gravitational dynamics and the microphysics of dark matter. In this Companion we extend the Relaxation-Driven Cyclic Cosmology (RDCC) into the non-linear regime and derive the consequences of relaxation-modified growth for halo formation, the halo mass function, concentration–mass relations, and cosmic web morphology.

Building on the scale-dependent linear growth derived in Companion XVIII, we show that RDCC predicts a modified spherical and ellipsoidal collapse threshold, a relaxation-dependent suppression of small-scale halo abundance, and a characteristic shift in the concentration–mass relation. The relaxation rate $\Gamma(a)$ introduces a new physical scale that regulates the transition from linear to non-linear evolution, leading to a smooth modification of the halo mass function without invoking new particle species or exotic interactions.

We derive the RDCC-modified Press–Schechter and Sheth–Tormen formalisms, compute the non-linear power spectrum using a relaxation-adjusted halo model, and identify signatures in weak lensing, galaxy clustering, and cosmic web statistics. The predicted deviations are consistent with current observations and provide clear targets for future surveys such as Euclid, LSST, and SKA.

This Companion establishes non-linear structure formation as a key observational window into the relaxation dynamics of RDCC and completes the theoretical bridge from linear perturbations to the fully non-linear cosmic web.

1 From Linear to Non-Linear Structure in RDCC

The transition from linear to non-linear structure formation in the Relaxation-Driven Cyclic Cosmology (RDCC) is governed by the same physical ingredients that generate scale-dependent growth in the linear regime (Companion XVIII): sector-dependent sound speeds, different free-streaming lengths, relaxation-induced damping, and anisotropic stress from inter-sectoral decoherence gradients (Companion XVII). These effects imprint a characteristic scale dependence on the linear matter power spectrum, which then propagates into the non-linear regime.

The linear RDCC matter power spectrum can be written as

$$P_{\text{lin,RDCC}}(k, a) = P_{\Lambda\text{CDM}}(k, a) \mathcal{S}_{\text{rel}}(k, a), \quad (1)$$

where the suppression function is

$$\mathcal{S}_{\text{rel}}(k, a) = \left[1 + \left(\frac{k}{k_{\text{rel}}(a)} \right)^2 \right]^{-\nu}, \quad \nu \approx 1-1.5. \quad (2)$$

Here,

$$k_{\text{rel}}(a) = \frac{\Gamma(a)}{c_{\text{eff}}(a)} \quad (3)$$

is the relaxation scale, with $\Gamma(a)$ the relaxation rate and $c_{\text{eff}}(a)$ the effective sound speed of the combined sectors.

1.1 Dimensionless Power Spectrum

We define the dimensionless power spectrum as

$$\Delta^2(k, a) = \frac{k^3}{2\pi^2} P(k, a). \quad (4)$$

In RDCC, the linear-to-nonlinear mapping becomes

$$\Delta_{\text{NL}}^2(k, a) = \mathcal{F}_{\text{RDCC}} \left[\Delta_{\text{lin}}^2(k, a) \right], \quad (5)$$

where $\mathcal{F}_{\text{RDCC}}$ is a relaxation-modified version of the standard halo-model mapping.

The most important consequences are:

- a smooth suppression of non-linear power on small scales,
- a turnover at $k \sim k_{\text{rel}}$,
- enhanced coherence on large scales,
- a smoother transition between linear and non-linear regimes.

1.2 Physical Origin of the RDCC Non-Linear Modifications

The non-linear regime inherits all scale-dependent features of the linear growth function $D(a, k)$:

- sector-dependent sound speeds suppress small-scale perturbations,
- free-streaming lengths differ between sectors,
- relaxation introduces friction-like damping,
- decoherence gradients generate anisotropic stress,
- the effective gravitational coupling becomes scale-dependent.

These effects modify:

- the collapse dynamics of overdensities,
- the spherical and ellipsoidal collapse thresholds,
- the halo mass function,
- the concentration–mass relation,
- the morphology of the cosmic web.

1.3 RDCC Non-Linear Mapping: Conceptual Overview

In standard Λ CDM, the mapping from linear to non-linear scales is approximately universal and scale-independent. In RDCC, however, the mapping becomes explicitly scale-dependent due to the presence of the relaxation scale k_{rel} .

We can write the RDCC non-linear mapping schematically as

$$\Delta_{\text{NL}}^2(k, a) = \Delta_{\text{NL}, \Lambda\text{CDM}}^2(k, a) \mathcal{M}_{\text{rel}}(k, a), \quad (6)$$

where the modification factor is

$$\mathcal{M}_{\text{rel}}(k, a) = \left[1 + \left(\frac{k}{k_{\text{rel}}(a)} \right)^2 \right]^{-\mu}, \quad \mu \approx 0.5-1. \quad (7)$$

This form captures:

- the smooth suppression of non-linear power,
- the delayed formation of low-mass halos,
- the reduced concentration of small halos,
- the enhanced vorticity near k_{rel} .

1.4 Summary

The transition from linear to non-linear structure formation in RDCC is governed by the relaxation scale $k_{\text{rel}}(a)$ and the scale-dependent linear growth function $D(a, k)$. These ingredients produce:

- a modified collapse threshold,
- a relaxation-adjusted halo mass function,
- a reduced concentration–mass relation,
- a suppressed non-linear power spectrum,
- characteristic changes in cosmic web morphology.

These effects form the foundation for the detailed non-linear predictions developed in the following sections.

2 RDCC-Modified Spherical and Ellipsoidal Collapse

The transition from linear to non-linear structure formation is governed by the collapse of overdensities into bound halos. In the standard Λ CDM model, this process is described by the spherical collapse model and its ellipsoidal generalization. In RDCC, however, the collapse dynamics are modified by the relaxation rate $\Gamma(a)$, the sector-dependent sound speeds $c_{s,i}$, and the relaxation-induced anisotropic stress derived in Companion XVII. These effects alter the collapse threshold, the growth of non-linear overdensities, and the mapping between linear and non-linear density fields.

2.1 Equation of Motion for a Spherical Overdensity

Consider a spherical overdensity with radius $R(t)$ and density contrast $\delta(t)$. The equation of motion becomes

$$\ddot{R} = -\frac{GM(< R)}{R^2} + \frac{1}{3}\Lambda R - \Gamma_{\text{eff}} \dot{R} + c_{\text{eff}}^2 \frac{k^2}{a^2} R, \quad (8)$$

where:

- $\Gamma_{\text{eff}}(a)$ is the effective relaxation damping,
- c_{eff}^2 is the effective sound speed of the combined sectors,
- the last term encodes the scale-dependent pressure-like contribution.

For a top-hat profile, the density contrast evolves as

$$\ddot{\delta} + (2H + \Gamma_{\text{eff}})\dot{\delta} = 4\pi G_{\text{eff}}(k, a)\rho_m \delta(1 + \delta) - \frac{4}{3} \frac{\dot{\delta}^2}{1 + \delta} + c_{\text{eff}}^2 \frac{k^2}{a^2} \delta. \quad (9)$$

2.2 Linearized Collapse Equation

Linearizing the above equation yields

$$\ddot{\delta}_L + (2H + \Gamma_{\text{eff}})\dot{\delta}_L = 4\pi G_{\text{eff}}(k, a)\rho_m \delta_L - c_{\text{eff}}^2 \frac{k^2}{a^2} \delta_L. \quad (10)$$

This equation shows explicitly that:

- relaxation damping slows collapse,
- the effective sound speed suppresses small-scale growth,
- the gravitational driving term weakens as $G_{\text{eff}}(k)$ decreases.

2.3 Scale-Dependent Collapse Threshold

The collapse threshold is defined as

$$\delta_c(k) = \delta_L(a_{\text{coll}}, k). \quad (11)$$

In RDCC, it becomes scale-dependent:

$$\delta_c(k) = \delta_{c, \Lambda\text{CDM}} \left[1 + \eta_1 \left(\frac{k}{k_{\text{rel}}} \right)^2 + \eta_2 \left(\frac{k}{k_{\text{rel}}} \right)^4 \right], \quad (12)$$

with typical values

$$\eta_1 \sim 0.05\text{--}0.1, \quad \eta_2 < \eta_1. \quad (13)$$

This is one of the central results of this Companion: **the collapse threshold increases on small scales**, delaying the formation of low-mass halos.

2.4 Virial Overdensity in RDCC

The virial overdensity is similarly modified:

$$\Delta_{\text{vir, RDCC}} = \Delta_{\text{vir, } \Lambda\text{CDM}} \left[1 - \chi \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (14)$$

with $\chi \sim 0.02\text{--}0.05$.

This reflects the fact that relaxation damping slows collapse and reduces the final virial density of small halos.

2.5 Ellipsoidal Collapse with Relaxation-Induced Anisotropic Stress

Ellipsoidal collapse is governed by the evolution of the three principal axes $a_i(t)$ and the tidal shear terms. In RDCC, the tidal field is modified by relaxation-induced anisotropic stress:

$$T_{ij}^{\text{RDCC}} = T_{ij}^{\Lambda\text{CDM}} + \Delta T_{ij}^{(\text{rel})}, \quad (15)$$

where

$$\Delta T_{ij}^{(\text{rel})} \propto \partial_{\langle i} \partial_{j \rangle} \Delta_{\text{dec}}. \quad (16)$$

This leads to:

- a mild increase in the ellipsoidal collapse threshold,
- reduced shear-driven collapse on small scales,
- modified Sheth–Tormen parameters.

The RDCC ellipsoidal collapse threshold becomes

$$\delta_{ec}(k) = \delta_c(k) \left[1 + \kappa \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (17)$$

with $\kappa \sim 0.1$.

2.6 Summary

RDCC modifies spherical and ellipsoidal collapse through:

- relaxation-induced damping,
- sector-dependent sound speeds,
- scale-dependent gravitational coupling,
- anisotropic stress from decoherence gradients.

These effects produce:

- a scale-dependent collapse threshold $\delta_c(k)$,
- a reduced virial overdensity,
- modified ellipsoidal collapse parameters,
- delayed formation of low-mass halos.

These results form the foundation for the RDCC halo mass function and non-linear power spectrum developed in the next section.

3 RDCC Halo Mass Function and Concentration–Mass Relation

The halo mass function (HMF) and the concentration–mass (c–M) relation are two of the most important non-linear observables in cosmology. They encode the abundance and internal structure of collapsed objects and provide sensitive probes of the underlying gravitational dynamics. In RDCC, the scale-dependent collapse threshold derived in Section 3 modifies both the HMF and the c–M relation in a characteristic and testable way.

3.1 Variance of the Smoothed Density Field

The variance of the smoothed density field in RDCC is

$$\sigma_{\text{RDCC}}^2(M) = \frac{1}{2\pi^2} \int_0^\infty k^2 P_{\text{RDCC}}(k) W^2(kR) dk, \quad (18)$$

where $R = (3M/4\pi\rho_m)^{1/3}$ and $W(kR)$ is the top-hat window function.

Using the RDCC suppression function,

$$P_{\text{RDCC}}(k) = P_{\Lambda\text{CDM}}(k) \left[1 + \left(\frac{k}{k_{\text{rel}}} \right)^2 \right]^{-\nu}, \quad (19)$$

we obtain the approximate scaling

$$\sigma_{\text{RDCC}}(M) = \sigma_{\Lambda\text{CDM}}(M) \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha}, \quad \alpha \approx \frac{2}{3}\nu. \quad (20)$$

3.2 RDCC-Modified Press–Schechter Mass Function

The Press–Schechter mass function becomes

$$\frac{dn_{\text{RDCC}}}{dM} = \sqrt{\frac{2}{\pi}} \frac{\rho_m}{M^2} \frac{\delta_c(k)}{\sigma_{\text{RDCC}}(M)} \left| \frac{d \ln \sigma_{\text{RDCC}}}{d \ln M} \right| \exp \left[-\frac{\delta_c^2(k)}{2\sigma_{\text{RDCC}}^2(M)} \right]. \quad (21)$$

Because $\delta_c(k)$ increases on small scales and $\sigma_{\text{RDCC}}(M)$ decreases, RDCC predicts:

- a smooth suppression of low-mass halos,
- unchanged abundance of massive halos,
- a turnover near M_{rel} .

3.3 RDCC-Modified Sheth–Tormen Mass Function

The ellipsoidal collapse threshold modifies the Sheth–Tormen parameters:

$$\delta_{ec}(k) = \delta_c(k) \left[1 + \kappa \left(\frac{k}{k_{\text{rel}}} \right)^2 \right]. \quad (22)$$

The RDCC Sheth–Tormen mass function becomes

$$\frac{dn_{\text{RDCC}}}{dM} = A \left[1 + \left(\frac{\sigma_{\text{RDCC}}^2}{a \delta_{ec}^2(k)} \right)^p \right] \frac{\rho_m}{M^2} \frac{\delta_{ec}(k)}{\sigma_{\text{RDCC}}(M)} \exp \left[-\frac{a \delta_{ec}^2(k)}{2\sigma_{\text{RDCC}}^2(M)} \right], \quad (23)$$

with mildly modified parameters a , p , and A .

3.4 RDCC Concentration–Mass Relation

Because relaxation delays collapse on small scales, halo concentrations are reduced:

$$c_{\text{RDCC}}(M) = c_{\Lambda\text{CDM}}(M) \left[1 - \chi \left(\frac{M_{\text{rel}}}{M} \right)^\beta \right], \quad (24)$$

with typical values $\chi \sim 0.05$ and $\beta \sim 2/3$.

This predicts:

- $c_{\text{RDCC}}(M) \approx (0.85\text{--}0.95) c_{\Lambda\text{CDM}}(M)$ for $M \lesssim 10^{12} M_\odot$,
- unchanged concentrations for massive halos.

4 RDCC Non-Linear Power Spectrum (Halo Model)

The non-linear matter power spectrum is written as

$$P_{\text{NL}}(k) = P_{1h}(k) + P_{2h}(k). \quad (25)$$

4.1 One-Halo Term

The 1-halo term becomes

$$P_{1h}(k) = \int dM \frac{dn_{\text{RDCC}}}{dM} \left(\frac{M}{\rho_m} \right)^2 |u(k|M)|^2, \quad (26)$$

where $u(k|M)$ is the Fourier transform of the halo density profile.

Because RDCC reduces concentrations, $P_{1h}(k)$ is suppressed for $k \gtrsim k_{\text{rel}}$.

4.2 Two-Halo Term

The 2-halo term becomes

$$P_{2h}(k) = \left[\int dM \frac{dn_{\text{RDCC}}}{dM} b_{\text{RDCC}}(M) u(k|M) \right]^2 P_{\text{lin,RDCC}}(k). \quad (27)$$

Because RDCC modifies both the bias and the linear spectrum, $P_{2h}(k)$ shows a mild, smooth suppression.

4.3 Combined RDCC Non-Linear Spectrum

The full RDCC non-linear spectrum exhibits:

- smooth suppression for $k \gtrsim k_{\text{rel}}$,
- unchanged large-scale clustering,
- reduced small-scale power consistent with weak lensing,
- no sharp cutoff (unlike warm dark matter).

5 Cosmic Web Morphology and Vorticity in RDCC

The cosmic web is shaped by the tidal field, velocity shear, and vorticity. In RDCC, all three are modified by relaxation-induced anisotropic stress.

5.1 Tidal Tensor Modification

The tidal tensor becomes

$$T_{ij}^{\text{RDCC}} = T_{ij}^{\Lambda\text{CDM}} + \Delta T_{ij}^{(\text{rel})}, \quad (28)$$

with

$$\Delta T_{ij}^{(\text{rel})} \propto \partial_{\langle i} \partial_{j \rangle} \Delta_{\text{dec}}. \quad (29)$$

This leads to:

- thicker filaments,
- larger voids,
- reduced connectivity of the cosmic web.

5.2 Vorticity Enhancement

Vorticity is generated at first order:

$$\vec{\omega} = \nabla \times \vec{v}, \quad \omega(k, a) \propto k^2 \Delta_{\text{dec}}(k). \quad (30)$$

RDCC predicts:

- enhanced vorticity near k_{rel} ,
- modified velocity-shear fields,
- observable signatures in kinetic SZ and peculiar velocity surveys.

6 Conclusion

In this Companion we have extended the Relaxation-Driven Cyclic Cosmology (RDCC) into the fully non-linear regime. Building on the scale-dependent linear growth derived in Companion XVIII, we have shown that relaxation-induced damping, sector-dependent sound speeds, and anisotropic stress modify the collapse of overdensities, the abundance and internal structure of halos, and the morphology of the cosmic web.

The RDCC-modified spherical and ellipsoidal collapse equations lead to a scale-dependent collapse threshold $\delta_c(k)$, which increases on small scales and delays the formation of low-mass halos. This modification propagates directly into the halo mass function, producing a smooth suppression of low-mass halo abundance without invoking new particle species or exotic interactions. The concentration–mass relation is similarly affected, with RDCC predicting reduced concentrations for halos below the relaxation mass scale M_{rel} .

Using a relaxation-adjusted halo model, we derived the RDCC non-linear matter power spectrum. The resulting suppression at $k \gtrsim k_{\text{rel}}$ is smooth, physically motivated, and consistent with the scale-dependent linear growth. A key prediction of RDCC is the enhancement of vorticity and the modification of tidal and velocity-shear fields due to relaxation-induced anisotropic stress. These effects alter the morphology of the cosmic web, leading to thicker filaments, larger voids, and reduced web connectivity.

Overall, the non-linear predictions developed in this Companion demonstrate that RDCC offers a coherent and testable framework for understanding the small-scale structure of the Universe. Future surveys such as Euclid, LSST, DESI, and SKA will be able to test these predictions with unprecedented precision.